

Gate 4 Cat 2 Substrate-Lie-Algebra Substrate-Pochhammer Schur II Ratio Candidate v0.1

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2026-06-04 Thursday ~12:58 EDT

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0. Goal

Per Lyra ALL 4 substrate-symplectic Cat 6 candidates walked back at substrate-canonical $Sp(2)$ substrate-fundamental representation level ($Pf(\omega) + \omega^2/2! + Tr$ ratio + Maslov) $\rightarrow$ substrate-mechanism source for Gate 4 honest open factor ~ 7 narrowed to Cat 2 substrate-Lie-algebra + Cat 3 substrate-Bergman categories. This v0.1 investigates Cat 2 substrate-Pochhammer Schur II ratio.

1. Schur II Cross-K-Type Substrate-Pochhammer

Per Lyra Mehler Matrix Element Framework v0.2 (Thursday ~10:00 EDT) + Wednesday Elie Toy 3742 + Helgason Ch. IX:

Schur II Pochhammer rule: substrate-K-type Mehler matrix element across substrate-K-types involves substrate-Pochhammer ratios per Sym^k representation theory.

Substrate-Pochhammer at substrate-spinor $V_{(3/2, 1/2)}$ vs substrate-fundamental $V_{(1/2, 1/2)}$ Sym^k ratio:

$(k+1)(k+2)/(2!)$ at substrate-spinor / $(1)(2)/(2!)$ at substrate-fundamental = $(k+1)(k+2)/2$.

For substrate-spinor $V_{(3/2, 1/2)}$ ($k=2$ from substrate-fundamental $V_{(1/2, 1/2)}$): - Substrate-spinor $(3+0) \cdot (3+1)/2! = 12/2 = 6 = C_2$ substrate-natural - Substrate-fundamental $(1+0) \cdot (1+1)/2! = 2/2 = 1$ substrate-natural

Substrate-Pochhammer ratio $V_{(3/2)}/V_{(1/2)} = 6/1 = C_2$ substrate-natural.

Wait — Wednesday Elie 3742 gave substrate-Pochhammer ratio = $20 = n_C \cdot 2^{\text{rank}}$, NOT 6. Let me re-examine.

2. Re-examination of Schur II Cross-K-Type Substrate-Pochhammer

Per Wednesday Elie Toy 3742: substrate-Pochhammer K-type ratio $\text{gen-2 } V_{(3/2, 1/2)} / \text{gen-1 } V_{(1/2, 1/2)} = 20$.

Per Sym^k representation of $Sp(2)$ substrate-fundamental V_{fund} dim 4: - Sym^2 of V_{fund} dim 4 has dim $(2+3)!/(2!3!) = 10$ - Sym^3 of V_{fund} dim 4 has dim $(3+3)!/(3!3!) = 20$

$V_{(3/2, 1/2)} \leftrightarrow Sym^3(V_{\text{fund}})$ per $Sp(2)$ substrate-symplectic representation theory; substrate-Pochhammer dim ratio $Sym^3/Sym^0 = 20/1 = 20$.

$20 = n_C \cdot 2^{\text{rank}} = 5 \cdot 4 = 20$ substrate-natural.

3. Substantive Substrate-Mechanism Reading

Substantive substrate-Pochhammer Schur II cross-K-type ratio $V_{(3/2, 1/2)} / V_{(1/2, 1/2)} = 20$ substrate-natural.

Per Gate 4 v0.1 honest open factor ~ 7 composition: substrate-mechanism contribution from Cat 2 substrate-Pochhammer cross-K-type ratio = 20 substrate-natural NOT $1/g = 1/7$ directly.

Substantive substrate-Pochhammer ratio 20 is in NUMERATOR of mass-mechanism composition (per Gate 4 v0.1: Pochhammer $\cdot$ Casimir $\cdot W_\lambda$) — NOT $1/g$ substrate-mechanism DENOMINATOR factor.

4. Cat 2 Substrate-Lie-Algebra Substrate-Mechanism Candidates for $1/g$

Per Cat 2 substrate-Lie-algebra substrate-mechanism source for $1/g$ substrate-natural factor:

(2a) Substrate-Lie-algebra substrate-Casimir RATIO $V_{(3/2, 1/2)} / V_{(1/2, 1/2)}$ at substrate-Casimir C_2 eigenvalue (2b) Substrate-Lie-algebra substrate-Pochhammer DENOMINATOR per Sym^k normalization at substrate-K-type (2c) Substrate-Lie-algebra substrate-Weyl ratio at substrate-K-type Weyl orbit substrate-natural form (2d) Substrate-Lie-algebra substrate-deficit per generation per Casey #13 Per-Generation Cluster Independence

(2a) Substrate-Casimir Ratio

Per Lyra Mehler Matrix Element v0.2: substrate-Casimir eigenvalues: $-V_{(1/2, 1/2)} C_2 = 5/2 - V_{(3/2, 1/2)} C_2 = 15/2$ - Ratio = $15/2 / 5/2 = 3 = N_c$ substrate-natural

$\sqrt{\text{Casimir ratio}} = \sqrt{3}$ substrate-natural. NOT $1/g$.

(2b) Substrate-Pochhammer Denominator

Per Sym^k normalization at substrate-K-type: substrate-Pochhammer denominator at $\text{Sym}^3(V_{\text{fund}}) = k!$ at $k=3$ = $6 = C_2$ substrate-natural.

Substrate-mechanism candidate: $1/k!$ at substrate-K-type $\text{Sym}^k = 1/6$ substrate-natural NOT $1/g = 1/7$.

(2c) Substrate-Weyl Orbit Substrate-Natural Form

Per Weyl orbit at substrate-K-type $V_{(\lambda_1, \lambda_2)}$: substrate-Weyl orbit substrate-natural form $|W(B_2)| = 8$ NOT $1/g$.

Substrate-Weyl substrate-orbit substrate-cardinality 8 NOT directly $1/g$ substrate-mechanism source.

(2d) Substrate-Deficit Per Generation per Casey #13

Per Casey #13 Per-Generation Cluster Independence: each generation has DISTINCT substrate-mechanism class.

Substantive substrate-deficit per generation candidate: gen-2 substrate-deficit = ? substrate-natural form per Casey #13 substrate-mechanism class.

Per Lyra Task #174 (Sunday): deficit = rank substrate-natural derivation. Substrate-deficit substrate-mechanism per generation may carry substrate-natural factor $1/g$ via substrate-genus substrate-mechanism cross-link.

Cat 2 (2d) substrate-deficit substrate-mechanism candidate: substantive multi-week per Cal #189 — substrate-deficit per generation substrate-mechanism could carry substrate-natural $1/g$ factor via substrate-genus substrate-mechanism layer.

5. Cat 3 Substrate-Bergman Substrate-Mechanism Candidates for $1/g$

Per Cat 3 substrate-Bergman / Shilov-boundary substrate-mechanism source for $1/g$ substrate-natural factor:

(3a) Substrate-Bergman kernel substrate-natural normalization $c_{FK} \cdot \pi^{(9/2)} = 225$ substrate-natural; substrate-mechanism contribution at substrate-Bergman matrix element level (3b) Substrate-Shilov-boundary substrate-codim-4 substrate-natural factor (per Casey #14 STANDING) (3c) Substrate-Bergman substrate-Liouville substrate-volume substrate-natural normalization per substrate-K-type

(3a) Substrate-Bergman Kernel $c_{FK} \cdot \pi^{(9/2)} = 225$

Per Sunday Elie Toy 3661 + Lyra Mehler Matrix Element v0.2: substrate-Bergman kernel substrate-natural normalization $c_{FK} \cdot \pi^{(9/2)} = 225 = 15^2 = (N_c \cdot n_C)^2$ substrate-natural.

Substrate-Bergman substrate-mechanism source for 1/g: substrate-Bergman kernel substrate-natural factor at gen-2 vs gen-1 substrate-K-type substrate-Bergman matrix element ratio.

Substantive multi-week: substrate-Bergman matrix element ratio $V_{(3/2, 1/2)} / V_{(1/2, 1/2)}$ per substrate-Bergman kernel substrate-natural normalization.

(3b) Substrate-Codim-4 per Casey #14 STANDING

Per Casey #14 STANDING (Thursday) + Lyra Gate 2 v0.2 Step 4.5-2 substantive: substrate-codim-4 substrate-natural factor from $1/n_C$ chirality projection + Casey #8 SCMP τ -direction substrate-mechanism.

Substantive observation: substrate-codim-4 substrate-mechanism per substrate-Shilov-boundary structure of D_{IV}^5 substrate. Substrate-mechanism factor 4 substrate-natural via codim-4 projection.

Substantive multi-week: substrate-codim-4 substrate-mechanism factor contribution to gen-2 $V_{(3/2, 1/2)}$ mass-mechanism composition; substrate-mechanism cross-link to Gate 4 honest open factor.

(3c) Substrate-Bergman Substrate-Volume Per Substrate-K-Type

Per substrate-Bergman substrate-Liouville substrate-volume substrate-natural normalization per substrate-K-type: substrate-Bergman substrate-volume ratio $V_{(3/2, 1/2)} / V_{(1/2, 1/2)}$ substrate-natural form.

Substantive multi-week: substrate-Bergman substrate-volume per substrate-K-type substrate-mechanism source for 1/g substrate-natural factor.

6. Closure v0.1

Cat 2 Substrate-Lie-Algebra + Cat 3 Substrate-Bergman substrate-mechanism candidate examination v0.1:

Cat 2 substrate-Lie-algebra: 4 sub-candidates examined — substrate-Pochhammer ratio = 20 (numerator), $\sqrt{\text{Casimir}}$ ratio = $\sqrt{3}$ (numerator), substrate-Weyl orbit = 8 (numerator), substrate-deficit per generation (multi-week candidate).

Cat 3 substrate-Bergman: 3 sub-candidates examined — substrate-Bergman kernel c_{FK} normalization 225 (multi-week), substrate-codim-4 Casey #14 STANDING (multi-week), substrate-Bergman substrate-volume per K-type (multi-week).

Substantive observation: Cat 2 substrate-Lie-algebra DIRECT substrate-invariants all in NUMERATOR (not 1/g denominator); substrate-mechanism source for 1/g substrate-natural factor most likely Cat 2 (2d) substrate-deficit per generation OR Cat 3 substrate-Bergman substrate-mechanism categories.

Refined Gate 4 honest open factor ~ 7 path: Cat 3 substrate-Bergman + Cat 2 (2d) substrate-deficit substrate-mechanism candidates load-bearing; multi-week explicit derivation per Cal #189.

Audit-chain Thursday afternoon — substantive substrate-mechanism source candidate-narrowing: substrate-symplectic Cat 6 walked back (4/4) → Cat 2 direct substrate-invariants in numerator (3/4 Cat 2 walked back at direct substrate-invariant level) → substrate-mechanism source narrows to Cat 3 substrate-Bergman + Cat 2 substrate-deficit substrate-mechanism categories.

Per Cal #36 STANDING positive-search rule + Cal #27 STANDING + Cal #189 question-shape: substantive examination → substantive narrowing → substrate-mechanism source for Gate 4 honest open factor ~ 7 = Cat 3 substrate-Bergman substrate-mechanism + Cat 2 substrate-deficit substrate-mechanism categories.

Tier: Cat 2 direct substrate-invariants walked back at substrate-canonical level (substrate-Pochhammer + $\sqrt{\text{Casimir}}$ + substrate-Weyl orbit all in numerator); Cat 2 (2d) substrate-deficit + Cat 3 substrate-Bergman

substrate-mechanism candidates multi-week per Cal #189; Gate 4 honest open factor ~7 substrate-mechanism source NARROWED.

— Lyra, Thu 2026-06-04 ~12:58 EDT. Cat 2 + Cat 3 substrate-mechanism candidate examination v0.1: 4 Cat 2 + 3 Cat 3 sub-candidates examined; Cat 2 direct substrate-invariants in numerator NOT 1/g denominator; substrate-mechanism source for Gate 4 honest open factor ~7 substrate-natural NARROWS to Cat 3 substrate-Bergman (c_FK normalization + substrate-codim-4 + substrate-Bergman substrate-volume per K-type) + Cat 2 (2d) substrate-deficit per generation candidates; multi-week per Cal #189.